Report(Titanic DataFrame with Pandas)

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1-Introduction:

What's my report about? Inclusive data analysis for a CSV file containing the real titanic data set, structured from just simple analysis(Exploration, trial and error, simple data cleaning & observation) to deeper analysis containing(EDA, discovering patterns, answering deep questions regarding the heart of the set, which is interesting how we can answer questions regarding patterns from such a huge DataFrame of size 891 rows , 12 columns. Such questions would need days to answer if we have a huge DataFrame. Though, with Pandas, a question can take only minutes.

Interested to know my key deductions and mysteries I've covered? Follow the next point:

2-Key Findings:

So, What secrets have I unfolded about titanic?

-The correlation between survival rate and the number of siblings or spouses a passengers might have(Could being responsible for siblings or a spouse risk your survival rate in Titanic's rules?)

-The average of survivors per each passenger class starting from the most luxurious to the least luxurious(1,2,3)

-The correlation between the survival rate and the class booked(could the class of a passenger be the reason for their survival ?!)

-The average number of survivors among men and women(Is there a gender gap when it comes to survival rate ? or are men and women equal ?)

-The standard deviation for the tickets' prices(Do the tickets' prices vary a lot or are the prices convergent to eachother?)

-The ratio of the total number of men to women(Which gender was more dominant in terms of count;Males or females?)

-The embarkation spot that has the highest survival Rate(Which embarkation had the highest number of alive passengers(C, S, or Q)?

-The combination of the Passenger Class and The Embarkation which had the highest survival rate among minors.

-The combination of the Passenger Class and The Embarkation which had the highest survival rate of the females of age from 21 to 50 with fare between 33.0000 and 79.0000.

Curious about the answer for all those questions? Get ready to get surprised in the next point.

3-Insights:

-- The correlation between survival rate and the number of siblings or spouses a passengers might have, is statistically valued by -0.035

Which means two things: -The relationship is weak since the modulus of it represents a very small value.

-The relationship is negative, hence incase It affects the passenger, It will be an inverse one, i.e the higher the siblings or spouses a passenger has, the less survival rate he has, but since our relationship is already weak, we can say having siblings or spouses doesn't affect the survival rate as much as other features.

-The average number of survivors per class shows that the class with the highest survival rate was among the passengers who booked the first class(The most expensive and luxurious one), which makes us think that the survival was kind of a function in one's ability to pay more. It doesn't feel fair, but it was Titanic's decision.

- The correlation between the survival rate and the class booked shows a negative relationship, which means that the higher the passenger class increases, the less the survival probability becomes.

-The survival rates between men and women show that the survival rates for women were much higher than the survival rates for men, where: The survival percentage for men= 18.9% approximately

The survival percentage for women=74.2% approximately

Which mathematically means that the average number of the men who survived is approximately quarter the actual mean of the women who survived!!!!!!! Big difference, which makes us remember the movie scene where men would sacrifice their lives in the sea during the destruction of Titanic to save their women.

-- The standard deviation for the tickets' prices(Fare column) is 49.7,

Which shows that the tickets' prices' range varies a lot in terms of the values inside Its range such that there are fares that are directed to the middle class people like (11.2) and others that are directed to people with high income such as (146.5).

--The ratio of the total number of men to women is 1.83(Greater than1) , hence the males are more dominant in terms of population than females, where: The number of males= 577 males

The number of females= 314 females

--The embarkation spot that has the highest survival Rate is (C) with a survival rate estimated at 55.4%

-- The combination of the Passenger Class and The Embarkation which had the highest survival rate among minors is (1, C), i.e The First Class and The Embarkation Spot (C) with a survival rate 100%

-- The combination of the Passenger Class and The Embarkation which had the highest survival rate of the females of age from 21 to 50 with fare between 33.0000 and 79.0000 is (1, C) (The First Class , Embark C) With a weighted score of 10).

Didn't you forget a very important question? Exactly! How did we output all these insights? What are the algorithms and methods we followed and went through ? That's what we'll discover in the last point.

4-Methodology:

Tools: ***Pandas(60%), Research(35%), Pure Python(5%)***

Algorithms and Steps:

1- ***Exploration:*** ***in real life, when you get exposed to a new situation like studying a new subject at school, you usually don't handle it right away. You keep looking at highlights of it first and explore it, so you can take a decision on how to initiate an action. Same thing goes with Data.***
   - ***Initially I had to look at the raw csv file of titanic dataset on Excel.***
   - ***Secondly, I started using coding by importing pandas as pd and making it read the file assigned to Variable, "titanic"***
   - ***Thirdly, I copied the original DataFrame and assigned it to a new variable called, "titanic1", so I don't affect the original DataFrame with my modifications.***
   - ***Finally, I used Pandas methods to explore by starting with head() to have a general look at the top of the dataset and try to recognize the values and columns generally, then I used info() to see important keys about the data set especially the names of all columns, their dtypes, the count of each dtype existing in the dataset, and the memory used by that set, so I can handle it perfectly with awareness in the data cleaning step. I used tail() to see the end of the DateFrame and then I felt the need to see unbiased results, so I ordered a random sample using sample(n=20, random_state=42). Lastly, I applied describe() to have raise my awareness of the statistical characteristics of the dataset like mean, standard deviation, quartiles, etc.***

I recognize your curiosity now about the reason for all what I've performed previously regarding all the exploration. The following method'll clarify everything.

2- ***Data Cleaning:*** ***After all the methods we've applied to the titanic data set to have a general look from the bird-level angle, we've drawn some conclusions, questions needing answers, insights need to be brought to light, specific patterns needing to be checked, and most importantly PROBLEMS preventing us from doing all that. Therefore, in this step, we'll try to clean out all the dust preventing us from seeing the dataset clearly.***

Handling the Data Types

- ***I started to notice from my exploration that there're dtypes that consume much memory and by looking at the values, I found that these values could accept a dtype consuming less memory.***

- ***I converted the dtypes of columns ["Survived", "Pclass", "SibSp", "Parch"] into Dtype int8 because the range of that dtype is -128 to 127, while the values of the mentioned columns are only between 1 and less than 50, hence that dtype is the most suitable for the values and memory usage.***

- ***I converted the dtype of Column "PassengerId" from int64 to category because I won't use it any in any calculations, so, I don't need the numeric form of it.***

  Why didn't I convert to str? Because str consumes much Memory, while category consumes the least memory usage.

- ***I converted Dtypes of Columns Cabin, Embarked, Ticket, Sex, and Name from str to category to save memory usage.***

- ***I converted Columns["Age", "Fare"] from float64 to float16 dtype to save memory .***

Removing & Dropping the troublemakers:

- ***After searching for duplicates using***

```
-   print(titanic1.duplicated().sum())
```

we found out there're no duplicates, so we'll search for another troublemakers

-By searching for the columns with nulls using

```
print(titanic1.isnull().sum())
```

we found that these are the only columns including nulls:

```
{Age:177 , Cabin: 687, Embarked:2} nulls
```

and we'll drop those nulls depending on the insights and the patterns we need to produce in EDA

3- EDA(Exploratory Data Analysis):

After looking for messy data and cleaning and organizing and sorting them , we start in this step to look for patterns, ask questions, answer them, draw conclusions, and insights which I've discussed in Part " Insights"

4- Save the new DataFrame (titanic1) with all the modifications we applied to serve satisfying our insights:

Using :

```
Titanic1.to_csv(r" D:\WUDownloadCache\titanic12.csv", index=False)
```

Done

By God's will